

From Leiden to Pleasure Island: The Constant Potts Model for Community Detection as a Hedonic Game

L.L. Felipe, K. Avrachenkov, D.S. Menasché

Nicholas C. Jacob

November 4, 2025

Clusters or Community

- Unsupervised Machine Learning Technique
- Networks
- Non-Overlapping Partitions
- Competing Objectives
 - Maximize Friends
 - Minimize Strangers
- Frustrated Choice (Cannot Max/Min both)

- A node is **robust** if its current community is unambiguously preferred over all other communities.
- For a given partition, the robustness will be the fraction of nodes that there is no better move for.

Familiarity Index

$$F_i(C_A, C_B) = \frac{\Delta d}{\Delta d + \Delta \hat{d}}$$

- $F > 1$ or $F < 0$ you should make that change
- $0 < F < 1$ are frustrated choices

$$v_{ij} = (A_{ij} - \gamma) \delta_{v_i, v_j \in C_k}$$

Can vary γ :

- $\gamma = 0$ ignores strangers
- $\gamma = 1$ ignores friends
- Frustrated nodes
 - $\gamma < F_i$ go to more friends
 - $\gamma > F_i$ go to less strangers